



RAMCO INSTITUTE OF TECHNOLOGY

**DEPARTMENT OF CIVIL
ENGINEERING**

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About the Department

- The Department of Civil Engineering was established in the year 2013. At present, it offers undergraduate programme, B.E. in Civil Engineering with a sanctioned intake of 60 students.
- The Department broadly covers important fields such as Surveying, Geotechnical Engineering, Structural Engineering, Hydraulics and Water Resources engineering, Construction Materials and Practices, Environmental Engineering and Transportation Engineering.
- The Department has well established modern laboratories to cater to the needs of academic and research activities. It is well equipped with computational facilities and resources both in terms of hardware and software.
- The Department constantly facilitates students from other departments of the Institute as a gesture of support to their academic, co-curricular and research desires.
- The Department has well-qualified and committed faculty members, lab technicians, and modern infrastructure in the form of elegant and smart classrooms, spacious laboratories, latest machinery and equipment. Students are encouraged to participate in co-curricular and extra-curricular activities to acquire leadership qualities.
- The Civil Engineering Association RIT STRUCTA has been functioning for the benefit of the students. It serves as a platform for students to make technical presentations, gain knowledge about recent advancements and bring together novel ideas. This Association aims at bridging the gap between the academic and the industry by inviting renowned industrialists and academicians to interact with students. Guest lectures, industrial visits, site visits, etc., are organized and value added courses are conducted on regular basis to ensure that students are conversant with recent developments in the field of Civil Engineering.
- The Department encompasses professional societies such as Indian Concrete Institute (ICI), ISTE chapters, which provides students a platform to acquire knowledge in various technical activities. It offers consultancy services in the fields of Structural Engineering, Geotechnical Engineering, Environmental Engineering, Surveying and Highway Engineering

DEPARTMENT OF CIVIL ENGINEERING

● VISION AND MISSION ●

VISION

create knowledgeable and professional competent civil engineers who are capable of providing global solutions to the problems related to civil engineering.

MISSION

- o Impart high quality education and technical skills to transform students into Professional Civil Engineers.
- o Establish state-of-the-art infrastructure for updating knowledge in recent developments and cutting edge technology.
- o Bridge the gap between academics and industry by practical training and providing solutions for design projects related to Civil Engineering.
- o Motivate the faculty members and students in the teaching learning process for achieving excellence to meet the global standards.
- o Produce graduates with high professional and ethical standards for fulfilling the needs of the industry and society.



"We shape our
buildings, ⁴
thereafter they
shape us"

Program Educational Objectives (PEOs)

Graduates of the programme will be professionally competent and apply the fundamental concepts of mathematics, science and Civil Engineering to solve problems in Civil Engineering.

Graduates of the programme will perform their responsibilities as professional Civil Engineers with quality and ethics.

Graduates of the programme, as part of an organization or as Entrepreneurs, will continue to learn with evolving technologies.

Graduates of the programme will innovate and provide global solutions for a sustainable built environment.

Graduates of the programme will contribute to the development of sustainable Infrastructure for the betterment of society.

Program Outcomes (POs)

Engineering Graduates will be able to:

- Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

A pull quote is an impactful quote taken from the article. You can place the quote you want to highlight here.

- The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

On completion of the degree, the students will be able to:

- Communicate and present civil engineering projects effectively.
- Use the techniques, skills, and modern engineering tools necessary for civil engineering practice and project management.
- Provide sustainable solutions to civil engineering problems.
- Perform as design consultants in construction industry for the design of civil engineering structures.

WHO IS THE GOOD ENGINEER?

Analytical thinking,
attention to detail,
creativity, communication
skills, leadership abilities,
time management,
technical knowledge,
problem-solving skills,
and adaptability.

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ARTICLES



Itaipu Dam

The Itaipu Dam is one of the world's largest hydroelectric power plants, located on the Paraná River along the border between Brazil and Paraguay. It is a binational project, managed by both countries through the Itaipú Binacional company. The dam is celebrated for its massive energy output and its role in providing clean, renewable energy to the region.



Itaipu Dam



Itaipu Dam



Key Features:

1. Size and Scale: Itaipu Dam is 7.9 kilometers (4.9 miles) long and 196 meters (643 feet) high. It has 20 generating units with a total installed capacity of 14,000 MW. The dam produces approximately 90% of Paraguay's electricity and around 10% of Brazil's electricity.

2. Construction: Construction began in 1975 and was completed in 1984. Building the dam required diverting the Paraná River, one of the largest river systems in South America. At its peak, over 40,000 workers were involved in its construction.

3. Energy Production: Itaipu is often mentioned alongside China's Three Gorges Dam as one of the highest electricity-producing dams globally. In 2016, it broke the world record for annual energy production by generating 103.1 million megawatt-hours (MWh) of electricity.

4. Environmental Impact: The Itaipu project has long been a point of environmental debate. On the one hand, it provides clean energy, but on the other hand, it submerged large areas of land, displacing thousands of people and altering ecosystems. However, Itaipú Binacional has invested in watershed restoration and environmental conservation projects, including reforestation efforts and the creation of wildlife corridors.

The Itaipu Dam is considered a modern engineering wonder and a critical piece of infrastructure for South America's energy grid.

Mrs .R. Kalaimani
AP/Civil

EXPLORING THE LATEST TRENDS AND TECHNOLOGIES IN GREEN BUILDING

Introduction:

In recent years, the global focus on sustainability and environmental consciousness has sparked a significant shift in the construction industry. Green building practices are gaining momentum as architects, engineers, and developers strive to create more eco-friendly and energy-efficient structures. This article delves into the latest trends and technologies in green building, highlighting innovative approaches that are revolutionizing the industry.



1.Net-zero energy buildings: Structures generate as much energy as they consume, effectively canceling out their carbon footprint. NZEBs achieve this through a combination of energy-efficient design, on-site renewable energy generation (such as solar panels), and energy storage systems. With advancements in solar power and energy storage technologies, net-zero energy buildings are becoming more attainable and economically viable.

2.Passive design and bioclimatic architecture: Passive design strategies leverage natural elements to optimize energy efficiency. Bioclimatic architecture takes into account local climate conditions, solar orientation, and natural ventilation to minimize the need for artificial heating, cooling, and lighting. Techniques such as designing for natural daylighting, incorporating green roofs and walls, and implementing intelligent shading systems help reduce energy consumption and enhance occupant comfort.

3.Green materials and sustainable construction: The use of eco-friendly materials is gaining significant traction in green building. Sustainable construction practices prioritize the selection of low-impact materials, such as recycled or reclaimed materials, rapidly renewable resources, and products with a reduced carbon footprint. Additionally, green building emphasizes waste reduction, recycling, and responsible sourcing of materials, promoting circular economy principles within the construction industry.

4.Smart building automation: Intelligent building automation systems are playing a vital role in optimizing energy performance and enhancing occupant comfort. These systems integrate various technologies, including sensors, advanced controls, and data analytics, to monitor and manage energy usage, indoor air quality, lighting, and HVAC systems. By continuously analyzing data and making real-time adjustments, smart building automation ensures optimal energy efficiency while providing a comfortable and healthy indoor environment.

5. Water conservation and management: Green building is increasingly focusing on water conservation and efficient management. Technologies such as rainwater harvesting, greywater recycling, and high-efficiency plumbing fixtures are being incorporated to reduce water consumption. Smart irrigation systems equipped with weather sensors and soil moisture monitoring help optimize outdoor water usage, preventing wastage and promoting sustainable landscaping practices.

6. Building-integrated renewable energy: The integration of renewable energy systems directly into the building envelope is gaining prominence. Building-integrated photovoltaics (BIPV) and solar thermal technologies enable the generation of clean energy while serving as architectural elements. Solar panels integrated into facades, roofs, or windows enhance the aesthetic appeal of buildings while offsetting energy consumption.

7. Internet of Things (IoT) and data analytics: The Internet of Things (IoT) and data analytics are revolutionizing the way green buildings are managed and optimized. IoT-enabled devices, such as occupancy sensors and smart meters, collect real-time data on energy usage, occupancy patterns, and indoor environmental conditions. Advanced analytics platforms analyze this data to identify energy-saving opportunities, optimize building operations, and provide valuable insights for ongoing performance improvements.

Conclusion: The green building movement is rapidly evolving, driven by a growing awareness of environmental sustainability and the desire to create healthier, more energy-efficient structures. The latest trends and technologies in green building, such as net-zero energy buildings, passive design, smart building automation, and building-integrated renewable energy, are transforming the construction industry. With ongoing innovation and increased adoption of sustainable practices, green buildings are set to become the norm rather than the exception, ushering in a more sustainable and resilient future.



Prepared by,
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HOW LOCOMOTIVES ARE CLASSIFIED BY INDIAN RAILWAYS

Locomotives can run under their own power or can run drawing power from elsewhere, converting fuel into pure kinetic energy that you can feel just by looking at it. And these are just two types of locomotives. There are many more types and identifying these locomotives is even a major sport across the World. In India, it is possible to know everything about a locomotive just by looking at it or by looking at a notation and it is no rocket science.

Gauge Power Load Class Subtype

W D M 3 A

Service Gauge	Traction Used	Traffic Type	Power/Version	Arbitrary
W Broad Gauge	D Diesel	P Passenger	Power Rating	Incremental
Y Meter Gauge	A AC Electric	G Goods	Diesels (-WDM2x)	Power Rating
Z Narrow Gauge	C DC Electric	M Mixed	Number x 1000hp	(Diesels Only)
N Narrower Gauge	CA AC+DC Dual	S Shunter	Version Number	Variants, Models
(Z = 2' 6" N = 2')	B Battery	U DMU/MEMU	Electrics+WDM2x	Variations (All)

The First Letter - The Gauge it Runs on.

The Second Letter - What Fuel it Uses.

The Third Letter - What it Hauls.

The Fourth Number - Version/Power

The Fifth Letter - The Subtype



Locomotives can be broadly classified into what fuel they use. But they can also be classified according to the gauge they run on. Then again, they can still be classified according to what work they are used for and according to their power. So which parameter do we use to classify locomotive types? Easy. We use all of them! To avoid confusion and to uniquely identify locomotives according to what they do and how they work, Indian Railways have developed a stunningly simple system which classifies locomotives into different classes taking into account all their parameters like Gauge, Traction, Usage, Version, Power etc.

Prepared by,
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Automated Highway System (AHS)

An Automated Highway System (AHS) is a proposed intelligent transportation infrastructure where vehicles are controlled using automated technologies to improve safety, traffic flow, and efficiency. The system involves the use of autonomous vehicles or semi-autonomous vehicles operating on specially designed highways with minimal or no human input.

Key features and components of an Automated Highway System:

1. Vehicle Automation

- Cars, trucks, and other vehicles are equipped with automation technologies such as sensors, cameras, GPS, and artificial intelligence (AI).
- These technologies enable vehicles to detect obstacles, maintain safe distances from other vehicles, follow traffic rules, and even respond to changing road conditions..

- Full automation would mean no human driver is required, and the system takes full control of acceleration, braking, and steering.

2. Dedicated Highway Lanes

- AHS often envisions dedicated lanes on highways that only autonomous vehicles can use. These lanes would have specialized infrastructure like sensors, communication systems, and enhanced road markings to support automation.
- The road network might also include systems that communicate directly with the vehicles (vehicle-to-infrastructure or V2I communication) for dynamic updates on traffic conditions or hazards.

3. Platooning

- Vehicles on the automated highway can travel in coordinated groups or "platoons." This reduces the space between vehicles, optimizing road usage and increasing fuel efficiency.

- Platooning allows for faster traffic flow and fewer bottlenecks since vehicles accelerate, decelerate, and steer in sync. Vehicles share data with each other and with road infrastructure to ensure seamless coordination and prevent accidents.

7. Energy Efficiency

4. Improved Traffic Flow

- With AHS, human errors that cause accidents, such as distracted driving or fatigue, would be significantly reduced.
- Automated control of speed and lane changes can smooth out traffic, reducing congestion, and improving overall travel times.

5. Safety

- The system is designed to reduce the risk of collisions by using real-time data to monitor traffic and road conditions.
- Technologies like adaptive cruise control, lane-keeping assistance, and automatic emergency braking would ensure safe operation even at high speeds.

6. Communication Technologies

- Vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication is vital to AHS.

- AHS can reduce fuel consumption by minimizing sudden braking and acceleration through smooth traffic flow and reducing aerodynamic drag when vehicles travel closely together.

Challenges of Implementing AHS:

- **Technological Limitations:** The required sensors, AI, and communication systems must be highly reliable, which is still a challenge in some environments (e.g., bad weather or complex urban areas).
- **Infrastructure Investment:** Significant upgrades are needed for highways, including communication systems, smart sensors, and road redesigns.
- **Legal and Regulatory Issues:** Implementing AHS requires new policies regarding safety standards, liability in accidents, and vehicle certification.

- Public Acceptance: Widespread acceptance of autonomous vehicles will depend on trust in the technology's safety and effectiveness.

Automated highways hold potential for transforming transportation, reducing traffic jams, increasing safety, and even minimizing environmental impacts through fuel savings. However, full implementation is still in the research and development phase.

PREPARED BY
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AP/CIVIL

Raised flooring



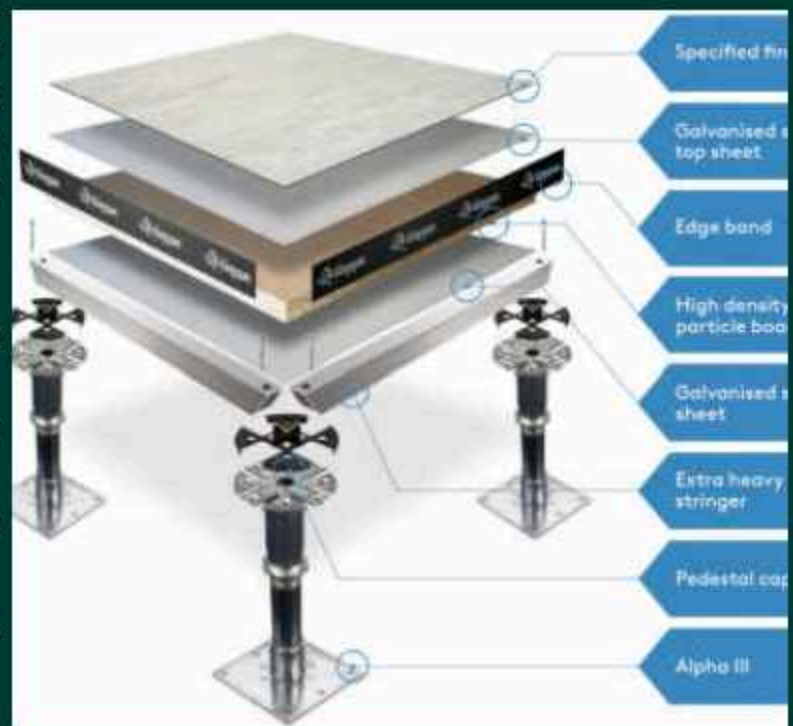
Raised flooring, access flooring, or raised access computer flooring, a raised floor is a construction model in which an elevated floor is built above a building's original concrete slab surface, thereby creating a hidden void between the two floors that is used for the passage of mechanical and electrical services.

There are two general types of raised floors:

- Traditional/Standard Access Floor (Post and Panel Raised Floor)
- Low Profile Access Floor (Cable Management Floor)

The type of raised floor needed depends upon several considerations:

- Application type (data center, office, medical facility, industrial site, etc.)
- Cooling system/air distribution setup
- Cabling/Wiring needs (flexibility, routing, distribution and delivery)



CURRENT APPLICATIONS OF THE RAISED ACCESS FLOOR:

- In today's world, raised access flooring is primarily used in IT data centers and computer rooms where there is a requirement for UFAD. All cabling and wiring are routed under the floor as well.
- A telecommunications facility is usually made up of continuous lineups of equipment cabinets. These data centers often have the most populated installation configuration in terms of square foot area, and, therefore, the heaviest floor load anticipated for a raised floor system.
- Moreover, data centers are often located in remote locations and are subject to physical and electrical stresses from sources such as fires and electrical faults.



- In data centers, isolated air-conditioning zones are often associated with raised floors. Perforated floor tiles are placed beneath computer systems to channel cool air directly into them. In turn, the computing equipment is designed to draw conditioned air from below into the room. Then an air conditioning unit draws the air from the room, cools it, and forces it back beneath the perforated tiles of the raised floor, thus completing the cycle.

Traditional post and panel raised floors are not used in a telecommunications/data center environment. Several different types of raised flooring systems can be employed instead and are bullet pointed below.

Stringerless raised floors: With this system an array of pedestals (posts) provide the necessary height for routing cables and also support each corner of the floor panels. No horizontal stringers are required.

Stringered raised floors: This system is composed of an array of steel pedestal assemblies (that consist of a steel base plate, a tubular upright, and a head) that are uniformly spaced on 2-foot centers and mechanically fastened to the concrete floor.

Structural platforms: The components of this system are constructed of steel angles or channels that are welded or bolted together to form an integrated platform used to support equipment.

Truss assemblies: This system utilizes attachment points to the subfloor to support a truss framework on which the floor panels rest. The truss has high lateral strength and transfers lateral loads to the subfloor with less strain than a vertical pedestal assembly does.

by:

Mrs. D. Darling Helen Lydia
AP/CIVIL

The Single Window Portal

The Single Window Portal for plan approval in Tamil Nadu is an initiative aimed at streamlining the process of obtaining approvals for building and construction projects. Here's an overview of its features and benefits:

Overview of the Single Window Portal

1. **Objective:** The portal aims to simplify and expedite the approval process for building plans, making it easier for developers, builders, and individual homeowners to obtain the necessary permissions.
2. **Integrated Services:** It provides a one-stop solution by integrating various departments involved in the approval process, such as local urban bodies, fire services, and environmental clearances.
3. **User-Friendly Interface:** The portal is designed to be accessible, with a user-friendly interface that allows applicants to submit their documents online and track the status of their applications.

Key Features

- **Online Application:** Users can fill out and submit applications online, reducing the need for physical visits to multiple offices.
- **Document Submission:** Required documents can be uploaded digitally, streamlining the process and minimizing paperwork.
- **Real-Time Tracking:** Applicants can track the status of their applications in real-time, providing transparency and reducing uncertainty.

- **Single Payment Gateway:** A consolidated payment system for various fees associated with plan approvals is available, simplifying financial transactions.

Benefits

1. **Time Efficiency:** The streamlined process significantly reduces the time taken to receive approvals, facilitating faster project commencement.
2. **Reduced Corruption:** By minimizing the need for physical interactions and paperwork, the portal aims to reduce opportunities for corruption and improve accountability.
3. **Improved Accessibility:** The online nature of the portal ensures that users from various regions can access services without the need to travel long distances.
4. **Enhanced Collaboration:** By integrating multiple departments, the portal fosters better communication and collaboration among various government agencies.

Mr .R. Muruganantham
AP/CIVIL

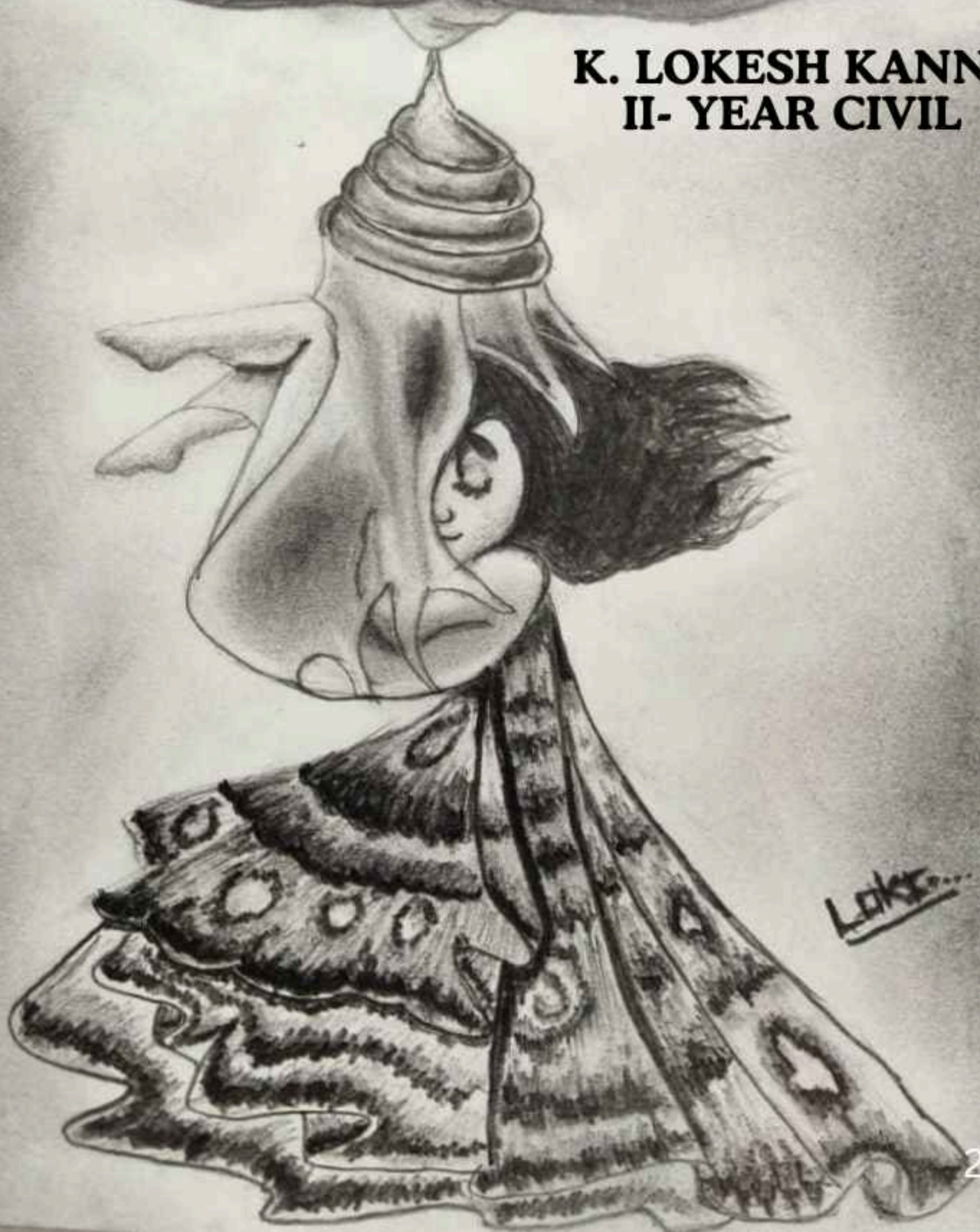
3

ART CORNER

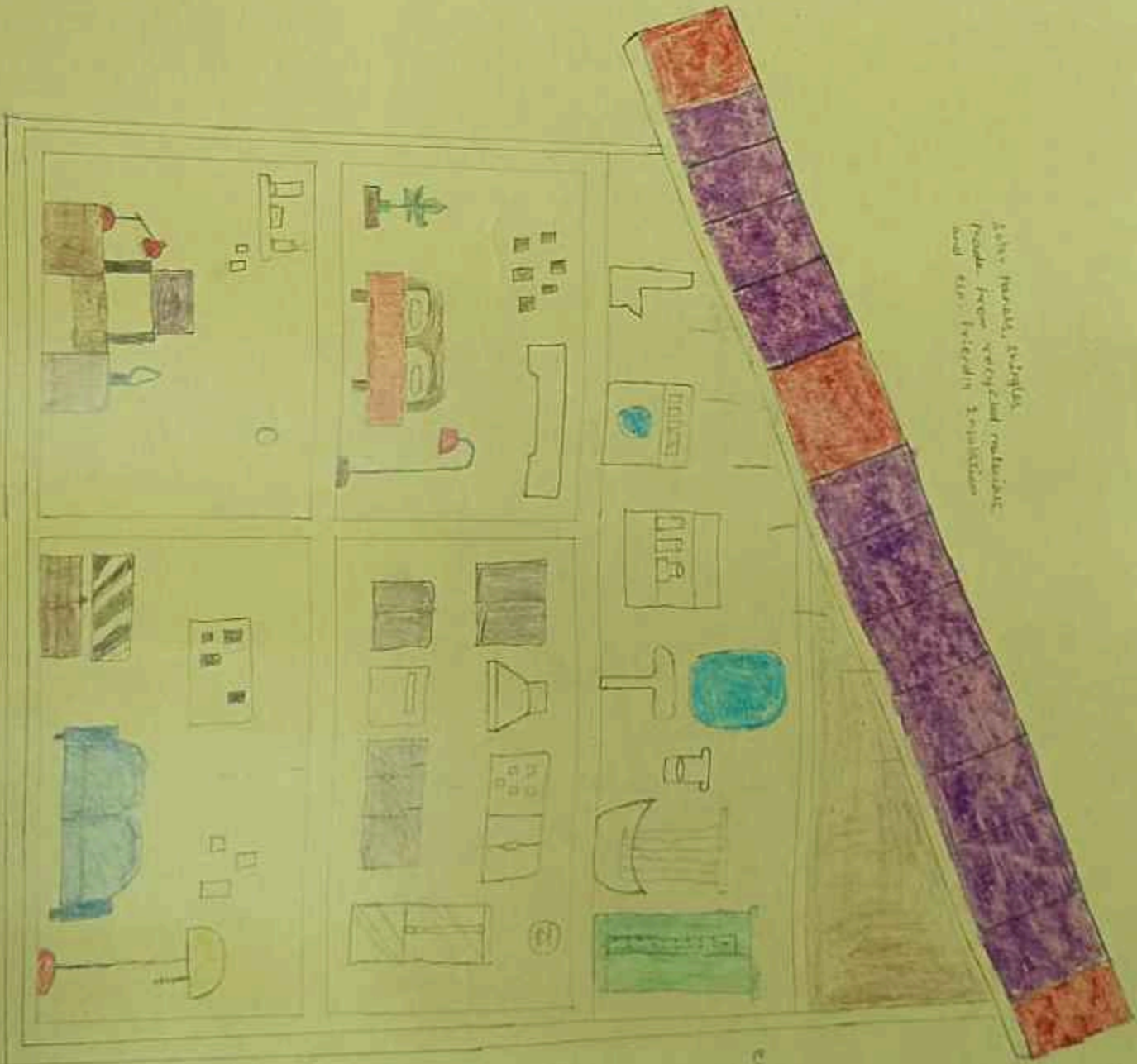
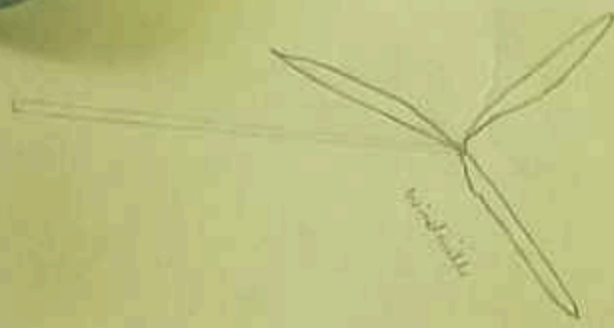




K. LOKESH KANNA
II- YEAR CIVIL



Surroundings
model
Built form



Also feasible, roofless,
made from recycled materials
and also friendly to nature

Sustainable structure
building materials like wood
utilized and houses floor
utilized

Efficient design building
800 natural light, and lower
climate control system

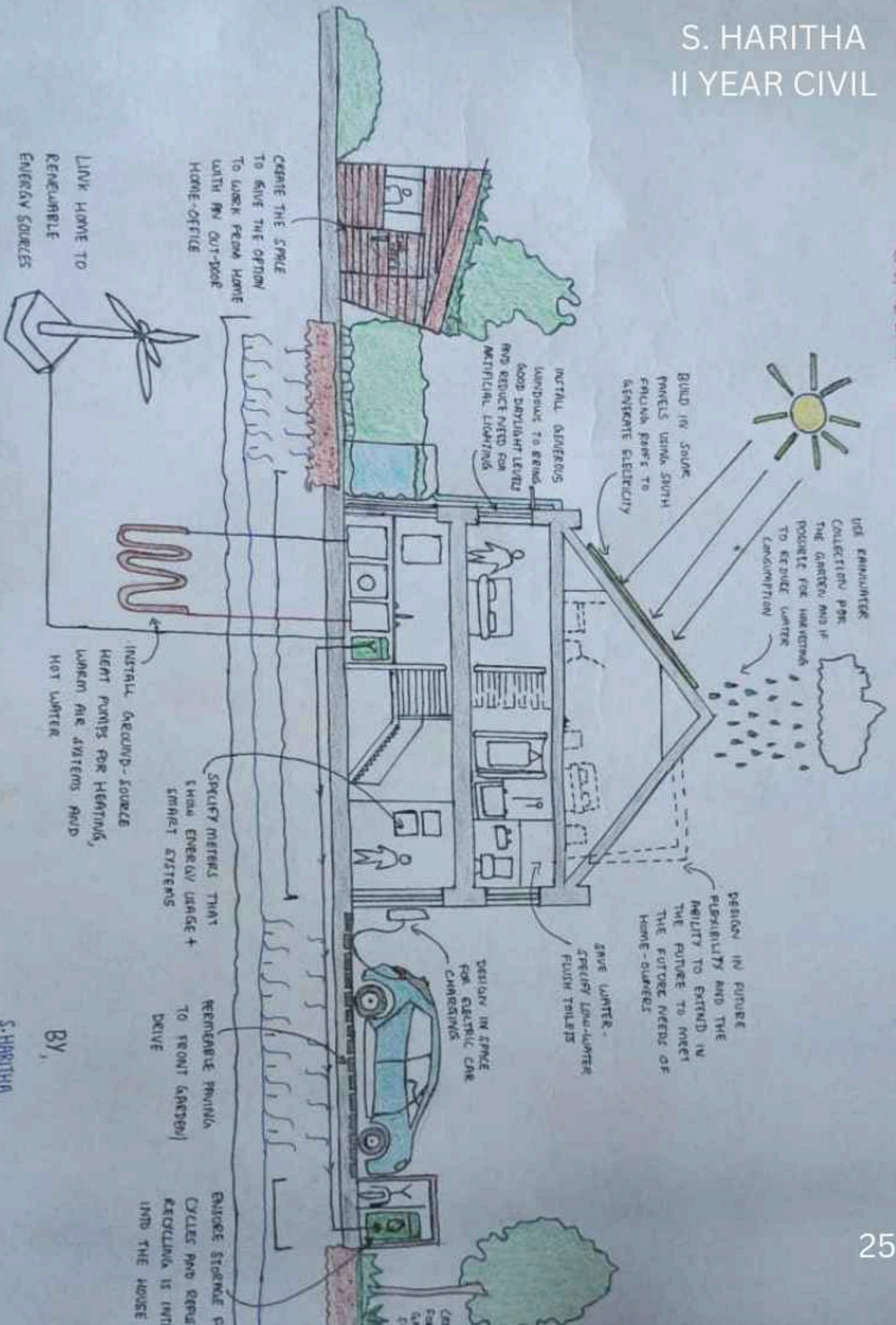
Energy saving system, modern,
flexible water supply, and
recycling

Indoor air circulation, water
conserving plumbing, noise
proof landscaping

M. Sasi Kumar
CIVIL 5th Year

S. HARITHA
II YEAR CIVIL

GREEN BUILDING CONCEPT AND SUSTAINABLE ARCHITECTURE



BY,
S. HARITHA,

SUSTAINABLE HOME BUILDING



low profile, design made
on recycled materials,
at eco-friendly locations.



flexible, reusable, building
materials are easy installed and
removal during.



efficient indoor cooling
system: natural ventilation
and cross ventilation system.



energy efficient windows,
insulating doors, screens, shades
and appliances.



Thermal insulation,
water conservation, planting
native plant landscaping.





Mr.S.Arun kumar
2nd year



Mrs. V S Priyadarshini
2nd year

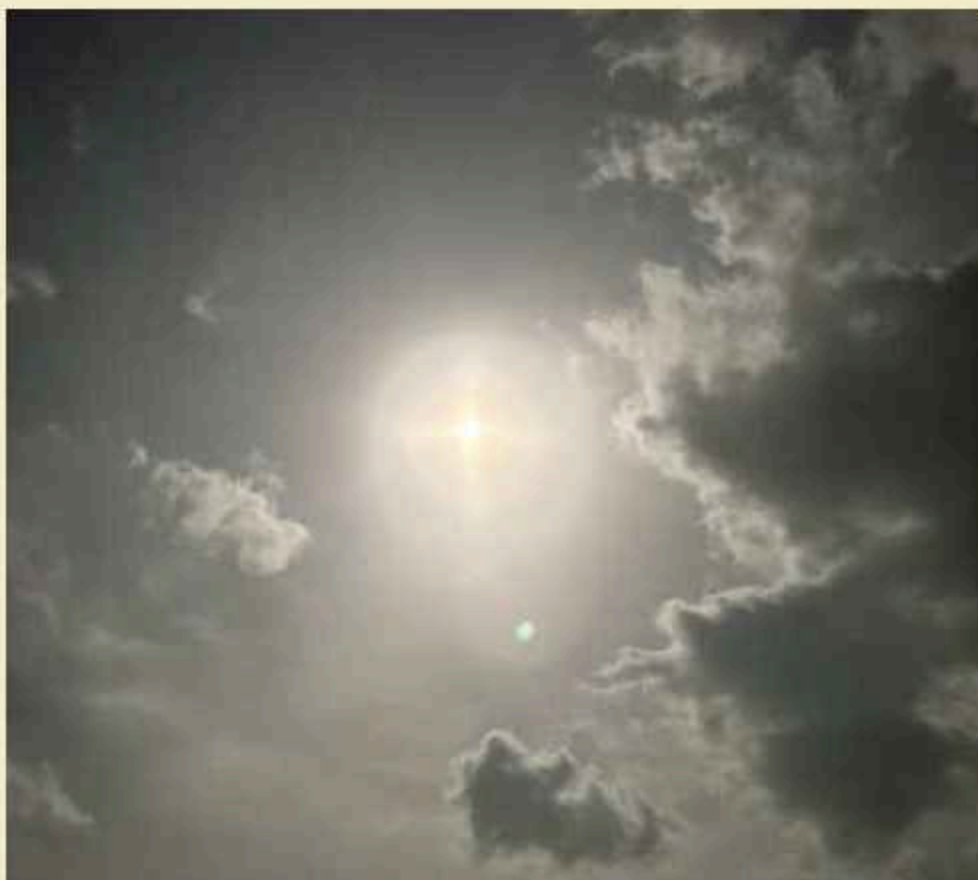


Ms.R.Aishwarya
2nd year





Mr.G.Mohan Rajan
4th year



Mr.G.Sivaramakrishnan
3rd year



Mr.P.Sankaralingam
3rd year